

Spring 2024

1) Write C++ program to print the series and also display corresponding sum for even numbers in the range from 2022 to 2300 for three iterations.

```
#include <iostream>
```

```
using namespace std;
```

```
main()
```

```
{  
    int start = 2022;
```

```
    int end = 2300;
```

```
    int sumEven;
```

```
    for (int i = 1; i <= 3; i++)
```

```
    {  
        sumEven = 0;
```

```
        cout << "i << " i << endl;
```

```
    }
```

```
    for (int num = start; num <= end; num++)
```

```
    {
```

```
        cout << num << "
```

```
        if (num % 2 == 0)
```

```
            sumEven = num + +;
```

```
    }
```

```
    cout << "sum of even numbers in iteration "
```

sumFive

```
    }
```

```
    return 0;
```

```
}
```

2) Write C++ program to calculate and print area and circumference of a circle. Take radius values from the user. Implement with arrays.

```
#include <iostream>
```

```
#include <math.h>
```

```
using namespace std;
```

```
main()
```

```
{
```

```
    int [5], size;
```

```
    double radius[5];
```

```
    double areas[5];
```

```
    double circumference[5];
```

```
    cout << "Enter size of radius:";
```

```
    for (int i=0; i<5; i++)
```

```
    {
```

```
        cout << "Radius " << endl;
```

```
        cin >> radius[i];
```

```
    }
```

```
    for (int i=0; i<5; i++)
```

```
    {
```

```
        area[i] = 2 * π * radius[i] * radius[i];
```

```
        circumference[i] = 2 * π * radius[i];
```

```
    }
```

```
cout << "Result\n"
```

```
    for (int i=0; i<5; i++)
```

```

}
cout << "Area" << area[i] << "\n";
cout << "Circumference" << circumference[i] << "\n";
}
return (0);
}

```

Write C++ program to calculate $Q = P^3 - 55P^2 - 37P + 46$ and print values for Q and P . Also find average of Q values. Take P values from the user for 10 iterations.

```
#include <iostream>
```

```
using namespace std;
```

```
main()
```

```
{
```

```
float Q, P, avg;
```

```
float sum = 0;
```

```
for (int i = 0; i < 10; i++)
```

```
{
```

```
cout << "Enter the value of P:"
```

```
cin >> P;
```

```
Q = P * P * P - 55 * P * P - 37 * P + 46
```

```
sum = sum + Q;
```

```
cout << P << "\t" << Q << endl;
```

```
}
```

```
avg = sum / 10;
```

```
cout << "Average =" << avg;
```

```
return(0);  
}
```

4) Write program to determine equivalent of n capacitors connected in series and parallel. Take capacitor values from the user.

```
#include <iostream>  
using namespace std;  
main( )  
{
```

write a program to calculate and print force value by taking input for mass and acceleration values from the user for 10 iterations.

```
#include <iostream>
```

```
using namespace std;
```

```
main()
```

```
{
```

```
double int mass, acceleration, force;
```

```
for (int = 1; i <= 10; i++)
```

```
{
```

```
cout << " i " << i << endl;
```

```
cout << "Enter mass:";
```

```
cin >> mass;
```

```
cout << "Enter acceleration:";
```

```
cin >> acceleration;
```

```
force = mass * acceleration;
```

```
cout << "Force = " << force << endl;
```

```
}
```

```
return(0);
```

```
}
```

Spring 2024 Long Q#2

```
#include <iostream>
```

```
#include <math.h>
```

```
using namespace std;
```

```
float f(float x)
```

```
{
```

```
    float k;
```

```
    k = 1 / x * x * x - 5;
```

```
    return(k);
```

```
}
```

```
main()
```

```
{
```

```
    float up, low, n, h, sum;
```

```
    cout << "Enter lower limit:";
```

```
    cin >> low;
```

```
    cout << "Enter upper limit:";
```

```
    cin >> up;
```

```
    cout << "Enter n:";
```

```
    cin >> n;
```

```
    h = (up - low) / n;
```

```
    cout << "step size is " << h;
```

```
    sum = f(low) + f(up);
```

```
    cout << "for Trapezoidal Rule:";
```

```
    for (int i = 1; i < n; i++)
```

```
        sum = sum + 2 * f(low + i * h);
```

```

cout << "for simpson 1/3:";
for (int i=1; i<n; i++)
{
    if (i%2==0)
        sum = sum + 2 * f(low + i * h);
    else
        sum = sum + 4 * f(low + i * h);
}
sum = sum * h / 3;

```

cout << sum << endl;

cout << "for simpson 3/8 rule:";

```

for (int i=1; i<n; i++)
{
    if (i%3==0)
        sum = sum + 2 * f(low + i * h);
    else
        sum = sum + 3 * f(low + i * h);
}

```

sum = 3 * h * sum / 8;

cout << sum << endl;

return 0;

}

```
2) #include <iostream>
using namespace std;
main()
```

```
(2) #include <iostream>
float udf (int a)
{
    float b;
    b = a / 1000.0;
}
main()
{
    float meter, kilometer;
    for (int i = 1; i <= 5; i++)
    {
        cout << "i" << i << ":" << endl;
        cout << "Enter distance in meters:";
        cin >> meter;
        kilometer = udf (meter);
        cout << "meter is equal to" << "kilomet
    }
    return (0);
}
```